PRELIMINARY EXAMINATIONS

HIGHER 1

TEMASEK JUNIOR COLLEGE

CANDIDATE NAME	
CIVICS GROUP	/
CENTER NUMBER	S
INDEX NUMBER	

CHEMISTRY **8873/02**

Paper 2 Structured Questions **24 August 2021**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Civics Group, centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
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2	
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Total	/ 80

This document consists of **26** printed pages and 2 blank page.

Section A

Answer **all** the questions in the spaces provided.

1 **(a)** Chlorine reacts with iodine to form a compound **T**, ICl_7. When dissolved in an excess of aqueous potassium iodide, **T** liberates iodine, I_2, which is the only iodine-containing product in the reaction.

(i) State the oxidation number of iodine in ICl_7. [1]

• +7

(ii) Write the equation for the reaction between **T** and potassium iodide. [1]

• $\mathbf{ICl_7 + 7KI \rightarrow 4I_2 + 7KCl}$

+7 7(-1) 0

(iii) Calculate the amount of iodine liberated when 1.00 g of **T** reacts with an excess of aqueous potassium iodide. [1]

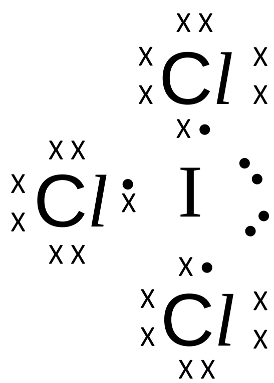
Amount of $\mathbf{ICl_7 = 1/(126.9 + 7 \times 35.5) = 2.66 \times 10^{-3}}$ **mol**

Amount of iodine $\mathbf{= 4 \times 2.66 \times 10^{-3}}$

• $\mathbf{= 1.06 \times 10^{-2}}$ **mol** (allow ecf from (ii))

(b) ICl_3 can also be formed from iodine and chlorine.

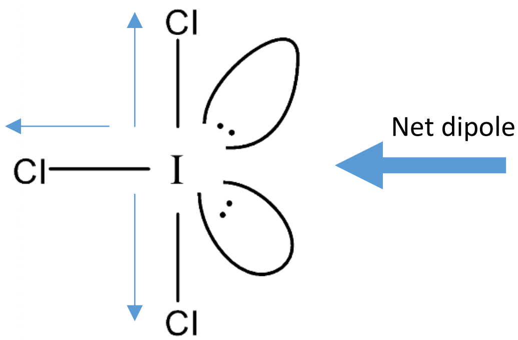
(i) Draw the dot-and-cross diagram for ICl_3. [1]



(ii) With reference to its molecular shape and partial charges, explain if ICl_3 is a polar molecule. [2]

Cl atom is more electronegative than I, hence Cl attracts the bonding electrons closer to itself. Thus, ✓Cl has δ- and I has δ+.

ICl_3 is a ✓polar molecule. ✓ICl_3 Is T-shaped and the dipoles of the polar I-Cl bonds do not cancel and there is ✓a net dipole moment in the molecule.



2 ✓ 1 mark

(c) Both chlorine and iodine can be formed from the decomposition of its respective hydrides.

State and explain the trend in the thermal stability of the Group 17 hydrides going down the group. [2]

- Thermal stability of HX **decreases down the Group.**

- **Bond energies decreases HF > HC*l* > HBr > HI. The H-X bond strength decreases** down the group and less energy is required to break the H-X bond. Thus HX is less thermally stable down the group.

(d) Sodium thiosulfate is a common reagent used for the reaction with iodine.

Write a balanced equation for the reaction between sodium thiosulfate and iodine and calculate the volume of 1.00 mol dm^{-3} sodium thiosulfate, in cm^3, required to react with all the iodine liberated in **(a)(iii)**. [2]

• $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

Volume of sodium thiosulfate required

= $(1.06 \times 10^{-2} \times 2)/1 \times 1000$

= • 21.3 cm^3

(allow ecf from (a)(iii))

[Total: 10]

2 **(a)** Excited states can be studied to gain information about the energies of orbitals that are unoccupied in an atom's ground state. The excited state of an element, **Q**, is represented by the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2 4p^1$.

(i) Identify the element **Q**. [1]

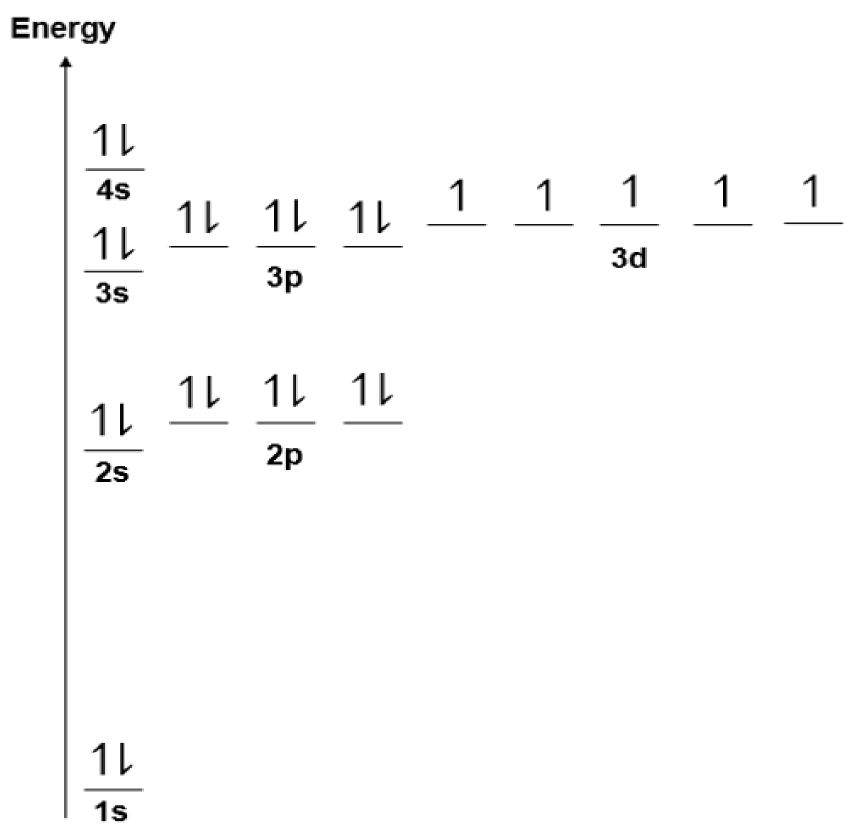
Since the total number of electrons is 25, • Q is manganese.

(ii) Draw the energy level diagram showing the electronic configuration of element **Q** in its ground state. [2]

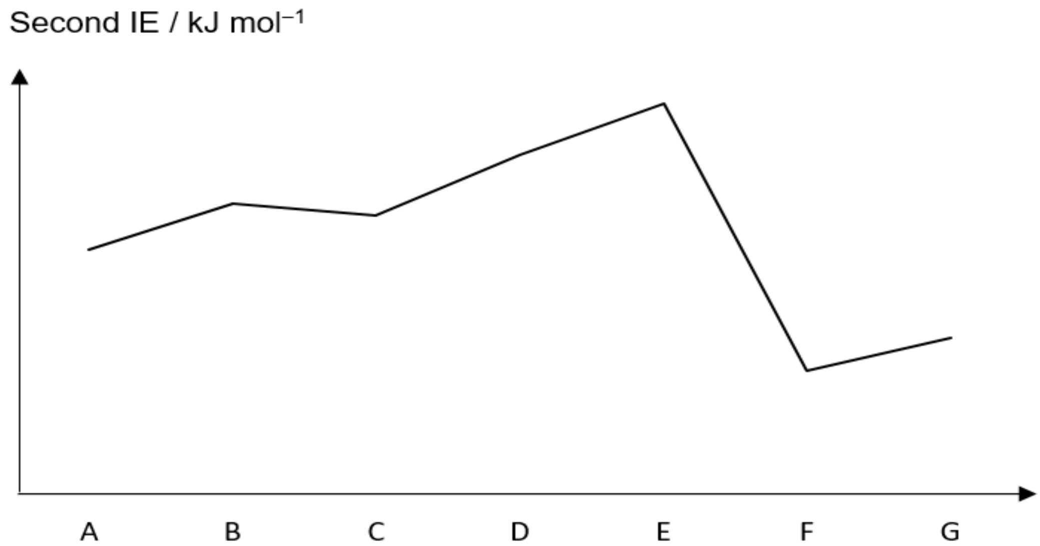
- **Electronic configuration of Q in ground state - $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$**

  (Either written out or shown correctly in energy level diagram)

- **Energy level diagram with axis and orbitals labelled, energy levels converge as distance from nucleus increase, energy diff between subshells < between shells.**



(b) The second ionisation energies of seven consecutive elements **A** to **G** in the Periodic Table are shown below.



(i) Define, with the aid of an equation, the second ionisation energy of oxygen. [2]

- **Minimum energy required to completely remove one mole of valence electrons from one mole of ground-state singly positively charged oxygen ions in gaseous state.**
- **O^+ (g) $\rightarrow$ O^{2+} (g) + e^-**

(ii) Explain the discontinuity in second ionisation energies between **E** and **F**, and between **B** and **C**. Hence deduce which element **A** to **G** is oxygen. [3]

- **There is a big drop in second IE from element E to F. Second electron in element F is removed from outer electronic shell. Element F is a Group 2 element (as it has two valence electrons).**

OR

There is a big drop in second IE from element E to F. Second electron in element E is removed from inner electronic shell. Element E is a Group 1 element (as it has one valence electron).

- **Hence element B is oxygen (Group 16).**
- **Second IE of C (Group 17) involves the removal of paired 2p electron which experiences inter-electronic repulsion. Hence, less energy is required to remove the paired electron from C^+ ion.**

(c) (i) State and explain the variation in bonding in Period 3 oxides in terms of electronegativity. [2]

- **Across the Period, electronegativity of elements increases, electronegativity differences between the elements and oxygen decreases resulted in sharing of electrons.**
- **The oxides become changes from ionic to covalent/increasingly covalent in character across Period 3.**

W, **X**, **Y**, and **Z** are four consecutive elements in the fourth period of the Periodic Table. The letters are not the actual symbols of the elements.

W forms an oxide that reacts with both acids and bases.

Z is a solid that can exist as several different allotropes. **Z** burns in air to form $\mathbf{Z}O_2$ which dissolves in water to form an acidic solution. This solution reacts with sodium hydroxide to form the salt $Na_2\mathbf{Z}O_3$.

(ii) Suggest the identities of **W** and **Z**. [1]

- **W is gallium, Z is selenium.**

(iii) Write equations for the reactions of oxide of **W** with sodium hydroxide and hydrochloric acid respectively. [2]

- $W_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaW(OH)_4$
- $W_2O_3 + 6HCl \rightarrow 2WCl_3 + 3H_2O$

*Accept equations that gave **W** as Ga.*

[Total: 13]

3 Chlorine is produced together with carbon monoxide when phosgene, $COCl_2$, undergoes dissociation according to the equation below:

$$COCl_2(g) \rightleftharpoons Cl_2(g) + CO(g)$$

The above reaction takes place in a 2 dm^3 reaction vessel. At the start of the reaction, there was 4.0 mol of phosgene in the vessel. When dynamic equilibrium was established, only 1.6 mol of phosgene was left.

(a) Write the K_c expression for the above reaction, stating its units. [1]

• $K_c = \frac{[Cl_2][CO]}{[COCl_2]}$ **Units: mol dm^{-3}**

(b) Calculate the equilibrium amounts of CO and Cl_2, and hence, determine the value of K_c for the above reaction, showing your working clearly. [2]

	$COCl_2$(g)	**⇌**	**Cl_2(g)**	**+**	**CO(g)**
Initial amount / mol	**4.0**		**0**		**0**
Change / mol	**- 2.4**		**+ 2.4**		**+ 2.4**
Eqm amount / mol	**1.6**		**2.4**		**2.4**

• **Equilibrium amounts of CO and Cl_2 = <u>2.4 mol</u> respectively**

• $K_c = \frac{(\frac{2.4}{2})(\frac{2.4}{2})}{(\frac{1.6}{2})}$ **= <u>1.80</u> mol dm^{-3}**

(c) State *Le Chatelier's Principle*. [1]

• ***Le Chatelier's Principle*** **states that when a <u>system at dynamic equilibrium</u> is subjected to a <u>change</u> which <u>disturbs the equilibrium</u>, the system will react in a way so as to <u>counteract the effect of the change to re-establish equilibrium.</u>**

(d) State and explain how the equilibrium composition might change if the above reaction is subjected to a decrease in pressure. [2]

By *Le Chatelier's Principle*, a decrease in pressure will cause the • <u>equilibrium position</u> to shift <u>right</u> to <u>increase the number of moles of gases</u> to <u>increase pressure</u>.
• The <u>new equilibrium mixture</u> contains <u>more products, Cl_2 and CO, and less reactant, $COCl_2$.</u>

[Total: 6]

4 **(a)** Nitrogen monoxide, NO(g), is a colourless, toxic gas that is formed by the oxidation of nitrogen. It can be reduced by hydrogen, $H_2(g)$, under certain conditions to form harmless products.

$$2NO(g) + 2H_2(g) \longrightarrow N_2(g) + 2H_2O(l)$$

(i) Define the term 'rate of reaction'. [1]

• Decrease (change) in reactant concentration per unit time or increase (change) in product concentration per unit time.

(ii) Identify a change in the reaction mixture that would allow the rate of reaction to be measured. [1]

• Decrease (change) in gas volume or pressure.

A series of experiments was conducted to determine the rate equation for the above reaction. The following data was obtained.

Experiment	Initial [NO] / mol dm^{-3}	Initial [H_2] / mol dm^{-3}	Initial Rate / mol dm^{-3} s^{-1}
1	2.50×10^{-3}	2.50×10^{-3}	1.27×10^{-3}
2	2.50×10^{-3}	4.60×10^{-3}	2.34×10^{-3}
3	5.00×10^{-4}	7.50×10^{-3}	1.52×10^{-4}

(iii) Determine the order of reaction with respect to NO and H_2. Hence state the rate equation. [3]

Using Expt 1 & 2

[NO] kept constant, [H_2] increases by $4.60 \times 10^{-3}/2.50 \times 10^{-3}$ = 1.84 times, rate increases by $2.34 \times 10^{-3}/1.27 \times 10^{-3}$ = 1.84 times.

- **First order wrt [H_2]**

Using Expt 1 & 3

Rate = k[NO]^y[H_2]

$1.27 \times 10^{-3}/1.52 \times 10^{-4} = k(2.50 \times 10^{-3})^y(2.50 \times 10^{-3}) / k(5.00 \times 10^{-4})^y(7.50 \times 10^{-3})$

- **y = 2, Second order wrt [NO]**
- **Rate = k[NO]^2[H_2]**

(iv) Using the data from Experiment 2, calculate the rate constant, k, stating its units clearly. [2]

Rate = k[NO]^2[H$_2$]

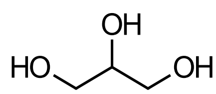
$2.34 \times 10^{-3} = k(2.50 \times 10^{-3})^2(4.60 \times 10^{-3})$

• $k = 8.14 \times 10^4$ • $mol^{-2}\ dm^6\ s^{-1}$

(b) Recently much interest has been shown in the production of the fuel biodiesel from algae. 50% of the mass of the dried algae are triglycerides. To convert triglycerides into biodiesel, the following processes are carried out.

$C_{17}H_{35}CO_2CH_2$
$C_{17}H_{35}CO_2CH$
$C_{17}H_{35}CO_2CH_2$

$\xrightarrow{\text{I}}$



$+\ 3C_{17}H_{35}CO_2H$

Triglyceride, M_r = 890 glycerol

$C_{17}H_{35}CO_2H \xrightarrow{\text{II}} C_{17}H_{35}CO_2CH_3$

Biodiesel, M_r = 298

(i) State the reactants and conditions for reactions I and II. [2]

Reaction I : • dil HCl, heat

Reaction II : • CH_3OH, conc.H_2SO_4, heat

(ii) Calculate the mass of biodiesel that can be produced from 1000 kg of dried algae. [2]

1000 kg algae contains 500 kg triglycerides

Amt of triglyceride = 500 x 1000/890 mol = 562 mol

• Amt of $C_{17}H_{35}CO_2H$ produced = 562 x 3 mol

= 1686 mol

= amt of biodiesel

• Mass of biodiesel = 1686 x 298 g = 502 kg

Alternative

• 890 g of triglyceride produces 3 × 298 = 894 g of biodiesel

1000 kg algae contains 500 kg triglycerides

• 500 kg produces 500 × 894/890 = 502 kg biodiesel

(iii) The production of biodiesel is at present an expensive process.

Suggest a reason why the development of biodiesel as an alternative to fossil fuels is important. [1]

Fossil fuels: causes serious environmental problems (global warming), imminent depletion

Biodiesel: renewable/sustainable, smaller C footprint (less global warming)

(c) Oxides of nitrogen are often produced in the exhaust fumes of car engines, together with other pollutants including hydrocarbons and carbon monoxide.

(i) Suggest another negative environmental consequence of the oxides of nitrogen. [1]

- **Nitrogen oxide can react with sulfur dioxide to form sulfur trioxide which dissolves in water to form acid rain.**

Or Nitrogen dioxide are harmful to vegetation eg leaf damage, decreasing growth.

(ii) Catalytic converters are used to reduce the emission of these harmful pollutants.

Write an equation for the reaction that occur in the catalytic converter that removes both carbon monoxide and nitrogen monoxide from the exhaust gases. [1]

• $\mathbf{CO + NO \rightarrow CO_2 + \frac{1}{2}N_2}$

(iii) Catalytic converters contain platinum nanoparticles.

Explain why platinum nanoparticles are used in the catalysis. [1]

• Nanoparticles have a large surface area to volume ratio making them effective catalysts.

(d) Human blood contains a buffer of carbonic acid, H_2CO_3, and bicarbonate anion, HCO_3^-, in order to maintain blood pH between 7.35 and 7.45. A value higher than 7.8 or lower than 6.8 can lead to death.

Explain, with the aid of equations, the role of carbonic acid and bicarbonate ion in controlling pH in blood. [3]

When the blood pH is too high

• $H_2CO_3\ (aq) + \mathbf{OH^-}\ (aq) \rightarrow HCO_3^-\ (aq) + H_2O\ (l)$

The H_2CO_3 neutralises the **excess OH^-, negligible change in pH** as the OH^- ions added are removed.

When there is excess acid i.e. blood pH is too low.

• $HCO_3^-\ (aq) + \mathbf{H^+}\ (aq) \rightarrow H_2CO_3\ (aq)$

The <u>HCO_3^- neutralises the</u> **<u>excess H^+, negligible change in pH</u>** as the H^+ ions added are removed.

- 1m for explanation

[Total: 18]

5 Nylon and polypropene are polymers that have a wide range of uses.

(a) Short chain alkenes is a feedstock for polymer and they can be obtained from the cracking of hydrocarbon chains.

Catalytic cracking is a process where long hydrocarbon chains are broken down into mixtures of molecules with shorter chains.

When an alkane $C_{11}H_{24}$, was subjected to catalytic cracking, three products were identified. The molecular mass of the mixture was 28, 42 and 44 with the relative proportions of 1: 2: 1 respectively. One of the products does not decolourise bromine.

Write a balanced equation for the catalytic cracking of $C_{11}H_{24}$. Show clearly the structural formulae of the three products and indicate the product that does not decolourise bromine.

[3]

There is conservation of atoms and only 3 products are formed.

$$C_{11}H_{24} \rightarrow \underset{M_r = 28}{CH_2{=}CH_2} + \underset{M_r = 42}{2CH_3CH{=}CH_2} + \underset{M_r = 44}{CH_3CH_2CH_3}$$

- 1m for balanced equation (Accept molecular formula)
- 1m for correct structural formula of the 3 products
- 1m for indicating the alkane $CH_3CH_2CH_3$ does not decolourise bromine.

(b) Apart from making containers, polypropene, also known as polypropylene, and nylon are used in making material for single use facemasks. Polypropene is used for the mask layers while nylon is used for the ear loops of masks.

According to WHO mask guide, masks are advised to have an inner most layer of absorbent, hydrophilic material, an outermost layer of hydrophobic material and a middle hydrophobic layer to enhance filtration or retain droplets. Masks should also meet the minimum requirements of filtration, breathability and fit.

Masks can also be made of woven or nonwoven fabric. The difference between the two fabrics is as shown in the table below.

	Nonwoven fabric	Woven fabric
Bonded by	Mechanical, glue or use of heat to melt fibers. Randomly oriented fibers are compressed.	Weaving yarns together via different construction patterns.
Properties	Light and cheap.	Washable and durable. Not stretchable unless elastic fibers are used.

(i) Suggest why the inner layer of the masks needs to be hydrophilic with high absorbance while the outer layer of the mask needs to be hydrophobic. [1]

• The inner layer needs to **absorbs water to trap droplets emitted by wearer**. The outermost layer of the mask **repels water or droplets emitted by others that may contain virus.**

(ii) With the knowledge of the bonding in polypropene, explain the suitability of using polypropene as the outer layer of single use masks [1]

• Polypropene is hydrophobic. It can **only form instantaneous dipole - induce dipole interactions** and **cannot form hydrogen bonds** with water hence it repels water.

(iii) Suggest two reasons why nonwoven polypropene is suitable in the manufacture of single use facemask. [2]

• **Easy to bond** / **can be quickly** produced by mechanical, glue or use of heat to manufacture the fabric/mask this allow **efficient production** / make more with less time.

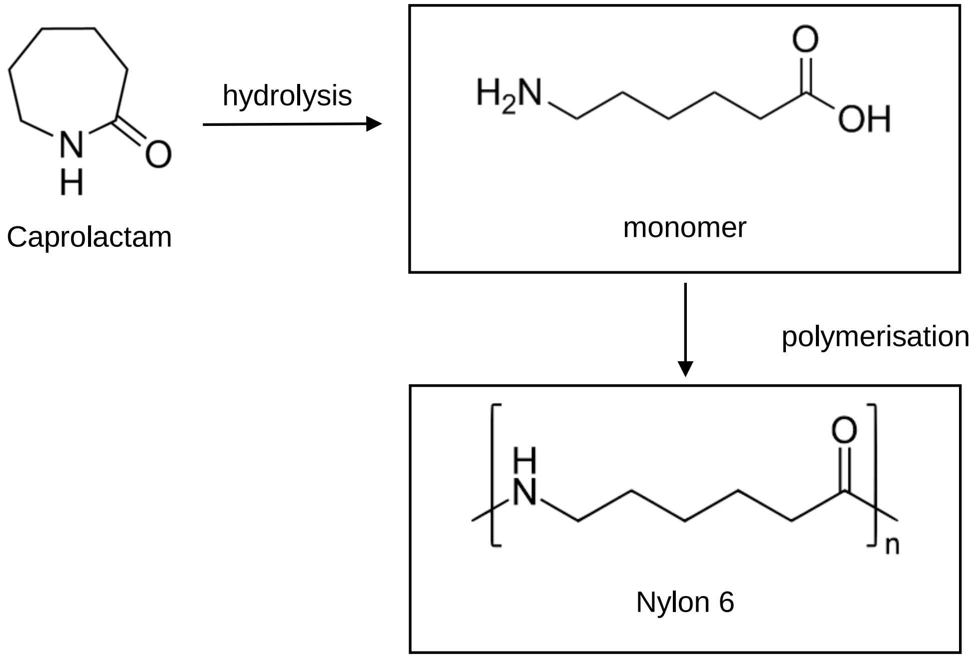
• It is **cheap** hence **production cost is low.**

• it is **light weight** hence **easy to transport**.

Accept other logical answers that links properties with the manufacture of facemasks.

(c) Nylon 6 is made via ring opening polymerisation of caprolactam. The reaction is initiated by the hydrolysis of Caprolactam and the subsequent polymerisation of the monomer.

(i) Draw the structure of the monomer and the polymer, Nylon 6 in the scheme below.



[2]

Some nylon fabric can be prone to creasing.

In order to remove creases in a nylon garment, an online blog proposed the following steps.

Step 1: Wet your wrinkled nylon garment thoroughly in lukewarm water. Wring out excess water.
Step 2: Place the wet nylon garment in the dryer. Turn the dryer to permanent press setting for synthetics (approximately 40°C) to prevent creases from forming during the drying process.
Step 3: Hang the nylon garment in a dry spot.
Step 4: Use a low heat iron to iron the nylon garment. Do not use high heat.

(ii) Suggest why wetting the nylon garment in Step 1 is necessary in removing creases in the nylon garment. [1]

• **Water will form hydrogen bonds** with existing amide linkages, **disrupting the existing hydrogen bonds between amide linkages**, which holds the creases in the garments.

(iii) Suggest why the use of low heat iron will remove creases in nylon garments. [1]

• It allows **hydrogen bonds between water and amide links to be broken** and **water to be removed or evaporate**. **Hydrogen bonds between amide linkages are reformed and reconnected** hence crease are removed.

(iv) Suggest why high heat iron may not suitable in removing creases. [1]

• Nylon is a thermoplastic, synthetic polymer. It has **a low melting point** and the garment **may melt under high temperatures**.

(v) Nylon polymers can be mixed with a wide variety of additives to achieve many different property variations. One method is the use of radiation to form cross links in Nylon.

Suggest how the use of additives to form cross-links changes the property of Nylon. [1]

The formation of cross-links will make the polymer become a thermosetting polymer. It will become **more rigid / have higher melting point / non-recyclable because it decompose on heating.**

• 1 mark for any of the property of thermosetting polymer that is different from thermoplastic.

[Total = 13]

Section B

Answer **one** question from this section in the spaces provided.

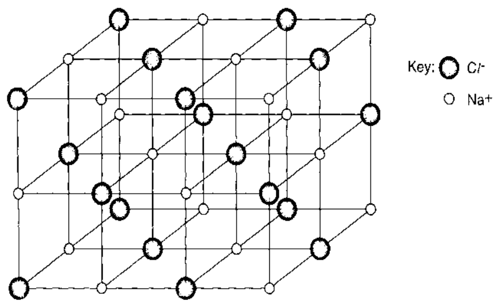
6 **(a)** Sodium chloride and iodine have lattice structures.

State the bonding in and describe the lattice structures of iodine and sodium chloride.

You may draw diagrams to illustrate your answer. [4]

NaCl have ✓ **<u>strong electrostatic forces of attraction (ionic bonds) between Na^+ and Cl^-.</u>**

It has ✓**<u>giant ionic (lattice) structure</u>** where the ions are arranged in a ✓ **<u>regular three-dimensional structure</u>** to maximise attractive forces between oppositely charged ions and minimise repulsive forces between similarly charged ions. In sodium chloride, ✓ **<u>each ion is surrounded by six oppositely charged ions</u>** (or shown in diagram)

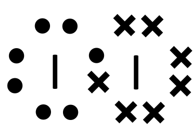


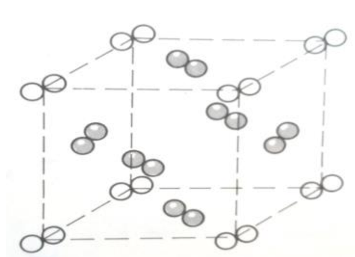
Crystal lattice of sodium chloride

In I_2, the ✓ **<u>strong electrostatic force of attraction between the localised shared electrons and the positive nuclei of the two I atoms constitutes the covalent bond</u>** which holds the 2 atoms together as a molecule. Or **<u>strong covalent bond between two I atoms in the molecule.</u>**

Note: use electrostatic force of attraction to define the covalent bond.

It has a ✓**<u>simple molecular structure</u>**, where the molecules are arranged in a ✓**<u>regular lattice structure (or shown in a diagram)</u>** ✓ **<u>held together by instantaneous dipole – induced dipole forces.</u>**





The I_2 molecules in the centre of each face are shaded.

2 ✓ 1 mark

(b) (i) With the aid of an equation, define *lattice energy of sodium oxide*. [1]

- **Lattice energy is the heat evolved when one mole of a solid ionic compound Na_2O is formed from its constituent gaseous ions, Na^+ and O^{2-} under standard conditions of 298 K and 1 bar.**

$2Na^+(g) + O^{2-}(g) \rightarrow Na_2O(s)$

(ii) The lattice energy of sodium oxide is –2720 kJ mol^{-1}.

Suggest whether sodium oxide or sodium chloride would have the more exothermic lattice energy. [2]

Lattice energy $\propto \left|\dfrac{q^+q^-}{r^+ + r^-}\right|$

- **Charge of O^{2-} > Cl^-**

  OR The magnitude of charge O^{2-} ions is twice the single charges on Cl^- ions.

  Radius of O^{2-} (0.140nm) < Cl^- (0.181 nm)

  OR The radii of O^{2-} ion (0.140nm) is smaller than the radii of Cl^- ion (0.181 nm).

- **Hence Na_2O has the more exothermic lattice energy.**

(iii) Lithium reacts less readily in air than sodium.

By reference to the Data Booklet, explain why lithium is less reactive than sodium. [2]

- **1st IE of Li = 519 kJ mol^{-1} 1st IE of Na = 494 kJ mol^{-1}**
- **Since the 1st IE of Li is larger than that of Na, more energy is required to remove the 1st electron from the valence shell of Li to form Li^+ ion as compared to ionising a Na atom to a Na^+ ion. As a result, it is relatively more difficult to form Li^+ and thus Li is less reactive than Na.**

OR

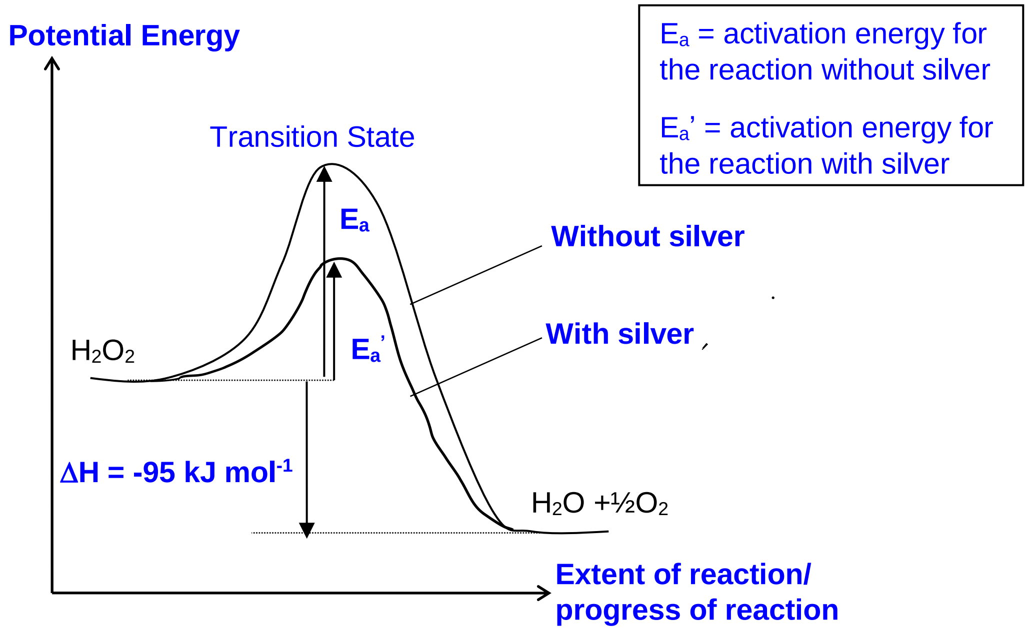
- **Radius of Li (0.152 nm) < Na (0.186 nm) [Not Ionic Radius]**
- **Since the atomic radius of Li is smaller than that of Na the valence electrons are more strongly held and less easily lost to form Li^+. Hence Li is less reactive than Na.**

(c) Hydrogen peroxide is an eco-friendly substitute for chlorinated agents in industrial whitening of cotton. It breaks down to water and oxygen, which are not harmful to the environment.

$$2H_2O_2 \rightarrow 2H_2O + O_2 \qquad \Delta H^{\ominus} = -95 \text{ kJ mol}^{-1}$$

This reaction can be speeded up by adding some silver powder.

(i) In the same diagram, construct a reaction pathway diagram for the decomposition of hydrogen peroxide with and without silver. [2]



- **Correct shape with labelling of reactants, products, E_a and ΔH**
- **Curve for reaction with silver shows a lower E_a, start and end at the same point as reaction without silver**

(ii) An enzyme, catalase, can be used to speed up the decomposition of hydrogen peroxide.

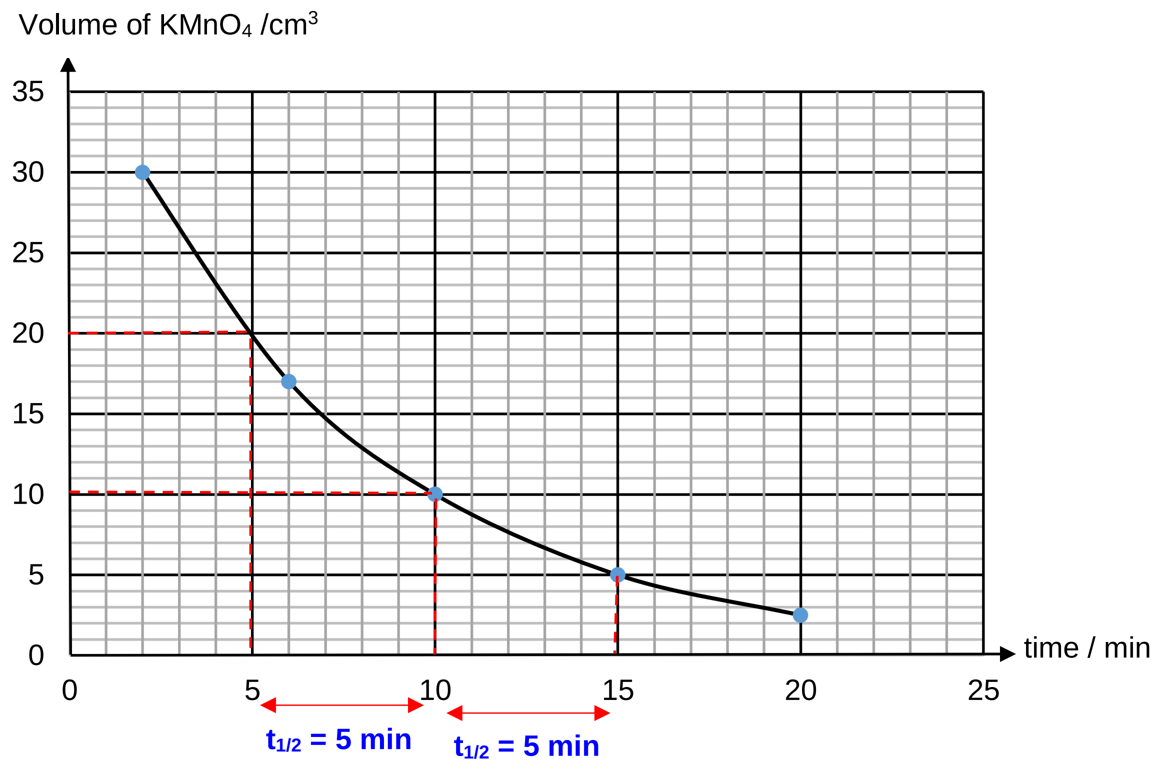
Papain is an enzyme that cleaves fibrous protein.

Explain the effect on the rate of decompositon of hydrogen peroxide if papain was used. [1]

- **Enzymes are <u>highly specific in its action</u>. Papain <u>cannot catalyse</u> the decomposition of hydrogen peroxide and the <u>rate would not increase</u>.**

(d) The kinetics of the decomposition of hydrogen peroxide in the presence of silver can be investigated by drawing portions of the solution at different time intervals, quenched and titrated with potassium mangante(VII).

A graph of volume of potassium manganate(VII) against time was then plotted.



(i) Deduce the order of reaction with respect to [H_2O_2]. [2]

- **It is <u>first order</u> with respect to [H_2O_2] as $t_{½}$ is <u>constant at 5 min</u>.**
- **Working with construction lines shown on graph**

(ii) Suggest and explain the effect on the rate constant for the decomposition of hydrogen peroxide in the absence of silver. [1]

- **The rate constant would be <u>smaller</u>. Without silver acting as a catalyst to <u>provide an alternative pathway with lower activation energy</u>, the <u>rate would decrease</u>.**

(e) Compound **Y** is a banned performance-enhancing stimulant. It may be prepared from compound **X** by the following three-step synthesis.

Compound **X** Compound **Y**

(i) Name compound **X**. [1]

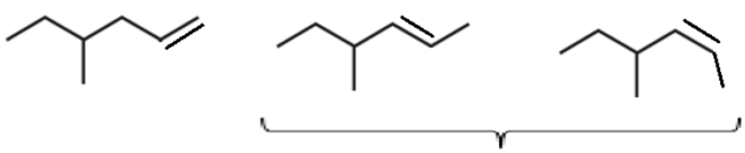
- **4-methylhexan-2-ol**

(ii) Suggest the type of reaction occurring in steps 1 and 3. [2]

step 1: • **oxidation**

step 3: • **reduction**

(iii) Compound **X** can react with hot Al_2O_3 to form a mixture of 3 isomers. Draw the structure of each isomer. [2]



(fyi : cis-trans isomers)

•• 2 marks for all 3 structures

[Total: 20]

7 **(a)** Hydrazine, N_2H_4, was first used as a component in rocket fuel in World War 2.

(i) Using the VSEPR theory, state and explain

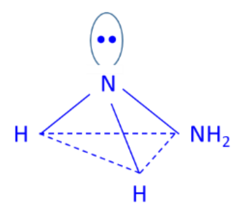
- the shape of hydrazine around each N atom;
- the H-N-H bond angle. [3]

There are ✓ 3 bond pairs and 1 lone pair of electrons in the outer/valence shell of each central N. By VSEPR theory, ✓ to maximise stability and minimise repulsion, the ✓ electron geometry is tetrahedral. Hence the shape is ✓ trigonal pyramidal.

Since ✓ lone pair – bond pair repulsion is greater than bond pair – bond pair repulsion, the H-N-H bond angle will be ✓ 107°.

Every 2 ✓ = 1 mark.

(ii) Hence, draw the structure of hydrazine, showing clearly the shape around one nitrogen atom. [1]



•

(iii) Like ammonia, hydrazine are bases.

Write an equation for the reaction between hydrazine and dilute hydrochloric acid. Identify the conjugate acid-base pair in the reaction. [2]

- **$N_2H_4 + HCl \rightarrow N_2H_5^+ + Cl^-$**
- **base: N_2H_4 conjugate acid: $N_2H_5^+$**

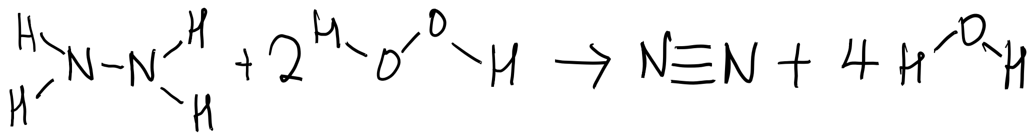
(b) When combined with hydrogen peroxide, the reaction produced nitrogen gas and water vapour, and released a large amount of energy that propelled the rocket to space.

$$N_2H_4 + 2H_2O_2 \rightarrow N_2 + 4H_2O \qquad \Delta H_r$$

(i) Define the term *bond energy* of a covalent bond. [1]

Bond energy is the • average energy required to break one mole of covalent bond in the gas phase into constituent gaseous atoms under standard conditions of 298 K and 1 bar.

(ii) Calculate the enthalpy change, ΔH_r, of the gaseous reaction between hydrazine and hydrogen peroxide. [2]



Bonds broken: 1 N-N, 4 N-H, 2 O-O, 4 O-H

Bonds formed: 1 N≡N, 8 O-H

Energy absorbed during bond breaking

= 160 + 4(390) + 2(150) + 4(460) = <u>3860</u> kJ

Energy released during bond forming = 944 + 8(460) = <u>4624</u> kJ

• correct energy released and absorbed

ΔH = 3860 + (-4624) = • <u>-764 kJ mol^{-1}</u>

(c) When two pure liquids, A and B, which do not react, are mixed together, heat may be absorbed or released depending on the relative strength of intermolecular forces of attraction formed in the mixture.

$$A(l) + B(l) \rightarrow A\text{---}B\ (l) \qquad \Delta H_{mix}$$

--- refers to the intermolecular forces of attraction between liquids A and B.

Enthalpy change of mixing, ΔH_{mix}, is the enthalpy change when 1 mole each of pure A and pure B are added to form a binary A---B solution.

The enthalpy change of mixing of ethyl ethanoate and trichloromethane is determined experimentally in an insulated calorimeter and the following data is collected.

	ethyl ethanoate	trichloromethane
Molar mass / g mol^{-1}	88.0	119.5
Mass / g	8.80	–
Specific heat capacity / J g^{-1} K^{-1}	1.92	0.96
Initial temperature / °C	31.2	31.2
Final temperature / °C	40.7	40.7

(i) Calculate the heat change of ethyl ethanoate during the mixing process. [1]

Heat evolved = $mc\Delta T$ = 8.8(1.92)(40.7 – 31.2) = • <u>161 J</u>

(ii) The heat change for an equimolar of trichloromethane at the same initial temperature is 109 J.

Calculate the heat change of the mixture and hence the enthalpy change of mixing, ΔH_{mix}, of ethyl ethanoate and trichloromethane. [2]

Total heat evolved = 161 + 109 = • <u>270 J</u>

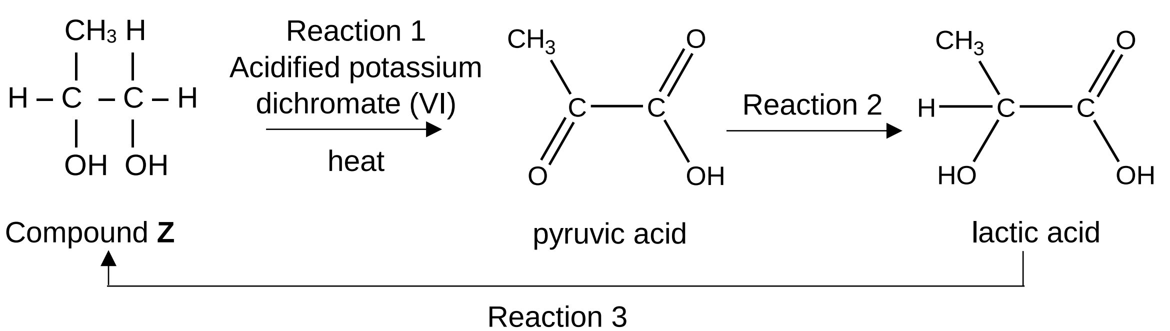
ΔH_{mix} = -270/(8.8/88) = • <u>-2.70 kJ mol^{-1}</u>

(iii) With reference to structure and bonding, explain the significance of the sign of ΔH_{mix} in **(b)(ii)** and hence, predict the observation of mixing ethyl ethanoate and trichloromethane. [3]

Both ethyl ethanoate and trichloromethane have ✓ <u>simple molecular structure</u>. ✓ <u>Formation of permanent dipole - permanent dipole between ethyl ethanoate and trichloromethane molecules</u> ✓ <u>release sufficient energy to break the</u> ✓ <u>permanent dipole - permanent dipole between individual molecules</u>. There is a net release of energy, so ΔH_{mix} < 0.

Hence, ethyl ethanoate and trichloromethane are • <u>miscible</u> and <u>form a homogeneous solution</u>. Every 2 ✓ = 1 mark

(d) Lactic acid can be formed from compound **Z** through the following reaction scheme.



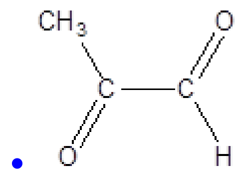
(i) State the reagents and conditions for reaction 2 and 3. [2]

Reaction 2 • $NaBH_4$ in alcohol, r.t.p

Reaction 3 • $LiAlH_4$ in dry ether, r.t.p

(ii) When Compound **Z** is warmed with acidified potassium dichromate (VI) with immediate distillation, another product is formed.

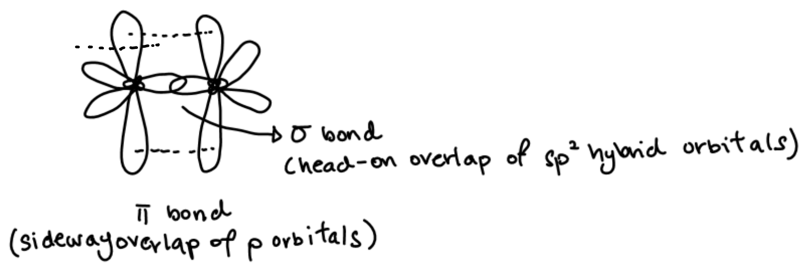
Draw the structure of the product formed. [1]



(iii) Compound **Z** can undergo a series of reactions to form an alkene.

Describe, in terms of orbital overlap, the bonding of the two carbon atoms of the C=C bond in alkenes. A clearly labelled diagram may clarify your answer. [2]

- C=C bond comprises of a sigma bond which arises from the head-on overlap of the sp^2 hybrid orbitals.
- A • pi bond from the sideway overlap of the unhybridised p orbital.



[Total: 20]